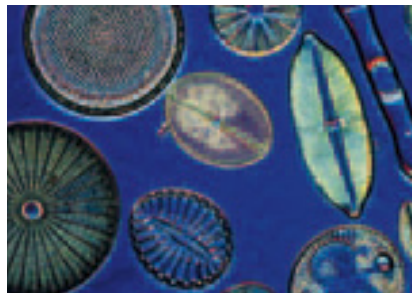


◀ AIRBORNE DUST

Huge quantities of airborne dust end up in the oceans and slowly settle through the water to add to the deep sediments forming the abyssal plains. A lot of dust is erupted from volcanoes, especially during catastrophic events like the 1991 eruption of Mount Pinatubo in the Philippines. The 1815 explosion of the Tambora volcano in Indonesia ejected so much dust that the sun was obscured for several months, causing the "year without a summer" in 1816. Much of this volcanic dust ended up in the oceans. Dust can also be blown off deserts, as shown in this satellite image of a storm carrying dust off the western Sahara and out over the Cape Verde Islands in the tropical Atlantic.

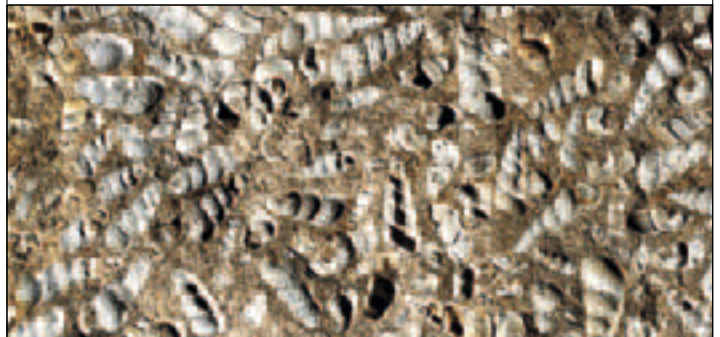
abyssal
plains



◀ BIOGENIC OOZE

Abyssal plains also consist of soft biogenic ooze, which is built up from the skeletal remains of microscopic marine organisms like these diatoms. But these dissolve in very deep water, so the main sediments on the deepest ocean floors are fine clays stained red by iron oxide.

SEDIMENTARY ROCKS



Over millions of years, ocean floor sediments are compressed into solid rock, which preserves in its characteristic layers a record of its formation. Many rocks have fossils of marine animals trapped in their layers, like these turret shells embedded in limestone. Each type of sediment turns into a different kind of rock: lime-rich oozes form limestone and chalk, sandy sediments form sandstone, and mud turns into shale. Dead organisms trapped within rock layers may not decompose properly, and through a long, complex process their carbon content can turn into the fossil fuels drilled from under the ocean floor by offshore oil and gas platforms.